REVIEW 3 • MACHINE LEARNING IMPLEMENTATION & DEFENSE

DiaRisk Analytics: Diabetes Risk Prediction and Model Evaluation

Review 3 — Machine Learning Implementation & Defense

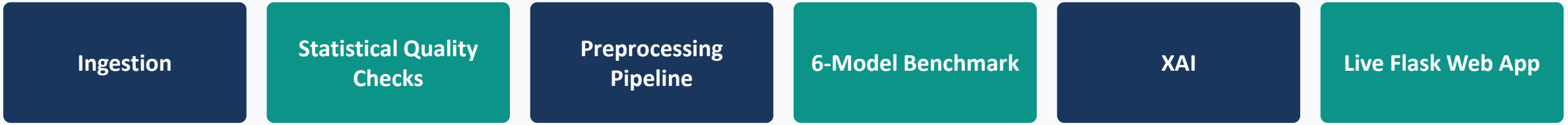
Course: Data Analytics and Visualization (24DSB3101) | AY 2026–2027

Team: Modalla Pranith Reddy (2420030158) • P.M. Pranitha (2420030277) • I. Hema Sri (2420030580)

Guide: Garimidi Siva Sree

Project Architecture & Dual-Dataset Framework

Problem: Early detection of diabetes risk using data analytics, machine learning, and Explainable AI (XAI).



Dataset 1 — CDC BRFSS 2015

- **229,781** clean records
- **21** survey features
- Multiclass target
- Severe **46.15:1** imbalance

Dataset 2 — Clinical & Lifestyle

- **97,297** records
- **28** lab / biometric features
- Binary target
- Balanced **1.50:1** ratio

1. Data Understanding — Dataset 1: CDC BRFSS 2015

229,781	21	46.15 : 1	3 Classes
Clean Rows	Features	Class Imbalance	Diabetes_012

Target: Diabetes_012

- **0** No Diabetes — **82.71%**
- **1** Prediabetes — **2.01%**
- **2** Diabetes — **15.27%**

Numerical & Binary

- **3 Continuous:** BMI (12–98), MentHlth (0–30 d), PhysHlth (0–30 d)
- **14 Binary (0/1):** HighBP, HighChol, CholCheck, Smoker, Stroke, HeartDiseaseorAttack, PhysActivity, Sex…

4 Ordinal Categorical

- **GenHlth** (1–5)
- **Age** (13 brackets: 1 = 18–24 … 13 = 80+)
- **Education** (1–6)
- **Income** (1–8)

1. Data Understanding — Dataset 2: Clinical & Lifestyle

97,297	28	1.50 : 1	Binary
Rows	Features	Balanced Target	diagnosed_diabetes

Target: diagnosed_diabetes — 0: Undiagnosed (**40.0%**) | 1: Diagnosed (**60.0%**)

8 Biochemical Labs

- **hba1c** (4.0–9.8%), **glucose_fasting** (60–172 mg/dL), **glucose_postprandial** (70–287)
- insulin_level; cholesterol panel (Total, HDL, LDL, Triglycerides)

5 Vitals & Biometrics

- **bmi** (15–39.2), waist_to_hip_ratio
- **systolic_bp** (90–179), diastolic_bp (50–110), heart_rate (40–105)

5 Demographic / Lifestyle (Continuous)

- Age (19–90), physical_activity (0–833 min/wk)
- diet_score, sleep_hours, screen_time

10 Categorical

- gender, ethnicity, education, income, smoking_status, alcohol_consumption, family_history_diabetes, hypertension_history

2. Data Cleaning & Pipeline Integrity

0
Raw Missing Values / Impossible Zeros

23,899
Duplicates Removed (9.42%)

2
Leaky Columns Dropped (Dataset 2)

Production-Grade Imputation (inside pipelines)

- **Numerical:** SimpleImputer(strategy='median') — robust to skewed clinical distributions
- **Categorical:** SimpleImputer(strategy='most_frequent') — preserves category integrity

Leakage Prevention

- **Deduplication** before splitting prevents data leakage
- **Target leakage:** dropped diabetes_stage and diabetes_risk_score to avoid circular training bias

3. Statistical Checks: Normality Distribution

Dataset	Feature	Skewness	Interpretation
Dataset 1	BMI	+1.21	Right-skewed (kurtosis = 2.65)
Dataset 1	MentHlth	+2.23	Severe right skew; zero-inflated, spike at 30 d
Dataset 1	PhysHlth	+1.98	Severe right skew; zero-inflated, spike at 30 d
Dataset 2	Age, BMI, Systolic BP, HbA1c	|skew| < 0.5	Approximately symmetric
Dataset 2	physical_activity	+1.14	Right-skewed
Dataset 2	insulin	+0.62	Right-skewed

Statistical Conclusion

Continuous features significantly depart from Gaussian normality (**$p < 0.001$**) — standard parametric tests are invalid.

3. Statistical Tests & Clinical Outlier Strategy

Non-Parametric Tests (alpha = 0.05)

- **Kruskal-Wallis (D1):** BMI H = 14,130.75; PhysHlth — p < 10^-300
- **Mann-Whitney U (D2):** hba1c U = 1.54e8, fasting glucose — p < 10^-300
- **Chi-square:** HighBP, HighChol, family_history — p < 10^-20

IQR Outliers	Feature	Share
Dataset 1	BMI / MentHlth / PhysHlth	3.88% / 14.27% / 16.14%
Dataset 2	hba1c (599)	0.62%
Dataset 2	glucose_fasting (722)	0.74%
Dataset 2	physical_activity	3.20%

DECISION: Retain All Outliers (No Truncation / No Deletion)

- **Clinical justification:** extreme values (HbA1c = 9.8%, fasting glucose = 172 mg/dL) are genuine disease pathology, not noise
- **Tree-based ensembles** are naturally robust to monotonic scaling and outliers

4. Data Preprocessing & Scaling Justification

Speaker 2: Pranitha

	Min-Max [0, 1]	StandardScaler (chosen)
Formula	(x − min) / (max − min)	z = (x − mean) / std
Outlier behaviour	One extreme value compresses 99% of patients into 0.10–0.25	Centres at mean 0, std 1; no squashing of normal variance
Optimisation	Distorted contours	Spherical loss contours for gradient & margin models

80 / 20
Stratified Split (random_state=42)

69 | 65
One-Hot Features (D1 | D2)

X_train only
ColumnTransformer fit — no leakage

- **Encoding:** OneHotEncoder(handle_unknown='ignore') removes false linear distance assumptions on ordinal survey codes
- **Pipeline:** Scikit-Learn ColumnTransformer fitted strictly on X_train to prevent test-set leakage

5. Model Building & Imbalance Strategy

1. Logistic Regression

- Linear probabilistic baseline; calibrated odds ratios

2. Decision Tree (CART)

- Non-linear hierarchical splits; mirrors clinical flowcharts

3. Random Forest

- Bagging of 100 de-correlated trees; reduces variance

4. SVM

- RBF kernel margin maximisation in high-dimensional space

5. XGBoost

- Gradient boosting, 2nd-order Taylor loss, L1/L2 regularisation

6. CatBoost

- Symmetric oblivious trees mitigate prediction shift

Imbalance Mitigation (Dataset 1, 46:1)

class_weight='balanced' and auto_class_weights='Balanced'

Model Persistence

Saved as complete .joblib pipeline objects under models/

6. Model Evaluation — Dataset 1 (CDC BRFSS 2015)

Model	Acc	Macro Prec	Macro Rec	Macro F1	ROC-AUC	5-Fold CV
XGBoost (Best)	**0.8363**	**0.4729**	**0.3914**	**0.4015**	**0.7656**	**0.7609 ± 0.0028**
Logistic Regression	0.6253	0.4466	0.5145	0.4243	0.7611	0.7557
Random Forest	0.6508	0.4389	0.5048	0.4278	0.7557	0.7455
CatBoost	0.6490	0.4388	0.5007	0.4260	0.7398	0.7348
SVM	0.4246	—	—	—	0.7388	0.7405
Decision Tree	0.5929	—	—	—	0.7216	0.7145

Confusion Matrix & Error Analysis (XGBoost)

- **TN = 37,030** | False Positive Rate **2.75%**
- Strong for non-diabetic screening
- Prediabetes (1.83% of data): near-zero recall without synthetic oversampling

Top Predictors (XGBoost)

- **HighBP** 23.13%
- **GenHlth** 14.80%
- **HighChol** 4.71% | **DiffWalk** 3.66%

6. Model Evaluation — Dataset 2 (Clinical & Lifestyle)

Model	Acc	Prec	Recall	F1	ROC-AUC	5-Fold CV
Decision Tree (Best)	**0.9195**	**0.9971**	**0.8685**	**0.9283**	**0.9450**	**0.9434 ± 0.0017**
CatBoost	0.9202	0.9995	0.8674	0.9288	0.9437	0.9430
Random Forest	0.9200	0.9990	0.8676	0.9287	0.9434	0.9438
XGBoost	0.9191	0.9970	0.8677	0.9279	0.9409	0.9414
Logistic Regression	0.8599	0.8733	0.8966	0.8848	0.9331	0.9337
SVM	0.8821	0.9320	0.8667	0.8982	0.9313	0.9345

Confusion Matrix (Decision Tree)

- TP = **10,141** | TN = **7,753**
- False Positives: only **30** (FPR **0.38%**)
- False Negatives: 1,536 (FNR 13.15%)

Clinical Validation (ADA Guidelines)

hba1c drives 95.01% of split decisions; fasting glucose **3.92%**

Directly validates ADA diagnostic cut-off: **HbA1c ≥ 6.5%**

Review 3 Summary & System Readiness

- ✓ **Data understanding:** Identified feature types and targets across both datasets
- ✓ **Data cleaning:** Imputed missing values, removed 23,899 duplicates, prevented target leakage
- ✓ **Statistical checks:** Validated non-normality; Kruskal-Wallis & Mann-Whitney; justified outlier retention
- ✓ **Preprocessing:** StandardScaler, one-hot encoding, ColumnTransformer pipelines
- ✓ **Model building:** 6 algorithms with balanced class weights; .joblib pipelines
- ✓ **Model analysis:** Metrics, rankings, confusion matrices, ADA clinical alignment

Full System Integration

- Live **Flask** application
- **http://127.0.0.1:5000**
- **MongoDB Atlas** prediction history
- **SHAP / LIME** Explainable AI